

3.2.2 What determines where particular species live?

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 5			
Plants often compete with each other for light and space, and for water and nutrients from the soil.	Students will not be required to recall any specific examples of nutrients. Suggested activity for TDA Compare the growth of plants when seeds are planted at different densities.	4.7.1.1	4.4.2.2
Animals often compete with each other for food, mates and territory.		4.7.1.1	4.4.2.2
Outcome 6			

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Animals and plants are subjected to environmental changes. Such changes may be caused by non-living or living factors.	An example of a non-living factor is a change in the average temperature or rainfall. An example of a living factor is the introduction of a competitor or predator. Suggested activity for TDA Compare the distribution of plants in a trodden and non-trodden area (using a transect line/quadrat).	4.7.1.2 and 4.7.1.3	4.4.2.3
Outcome 7			
Pollution of the environment can occur: • in water, from sewage, fertiliser or toxic chemicals • in air, from smoke and gases such as sulfur dioxide which contributes to acid rain • on land, from landfill and from toxic chemicals such as pesticides and herbicides, which may be washed from land into water. Students should recognise that rapid growth in human population means that more resources are used and more waste is produced.	Suggested activity for TDA Investigate whether rainwater in a city is more acidic than rainwater in the countryside. Suggested activity for TDA Compare the quality of water from different sources eg running water and still water.	4.7.3.2	4.4.2.6